

Bangabandhu Sheikh Mujibur Rahman University, Kishoreganj
Department of Computer Science and Engineering
1st Year 2nd Semester B.Sc. (Engg.) Final Examination, Session: 2021-22
CSE 1202: Digital Logic Design Lab (1.5 Credits)

Time: 90 Minutes

SET-A

Full Marks: 20

INSTRUCTIONS:

- a. Use + line of the breadboard for shorting **5V** and – line for shorting **GND**.
- b. Use a minimal number of **ICs** and **wires** for the hardware implementation.
- c. **Follow the orientation of the higher significant bit to lower significant bit from left to right or up to down.**

Question-1

Marks

20

A program in C language performs as mentioned below.

The program takes two integers P and Q as input and displays the value of $P + Q$ if $P < Q$, otherwise displays the value of $P - Q$.

Unfortunately, for today you will have to do either the hardware implementation or the Proteus implementation of the program where you can safely assume that

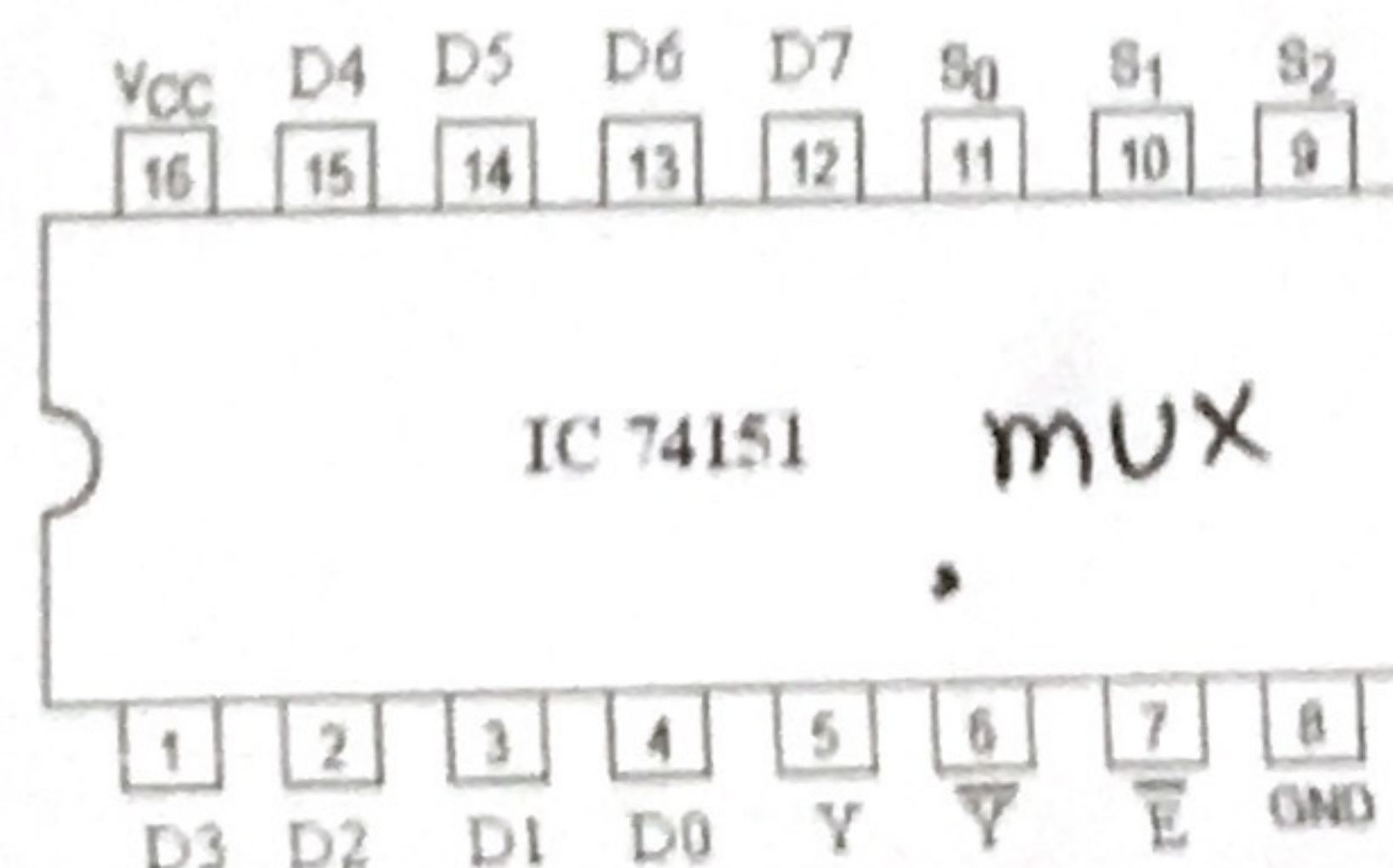
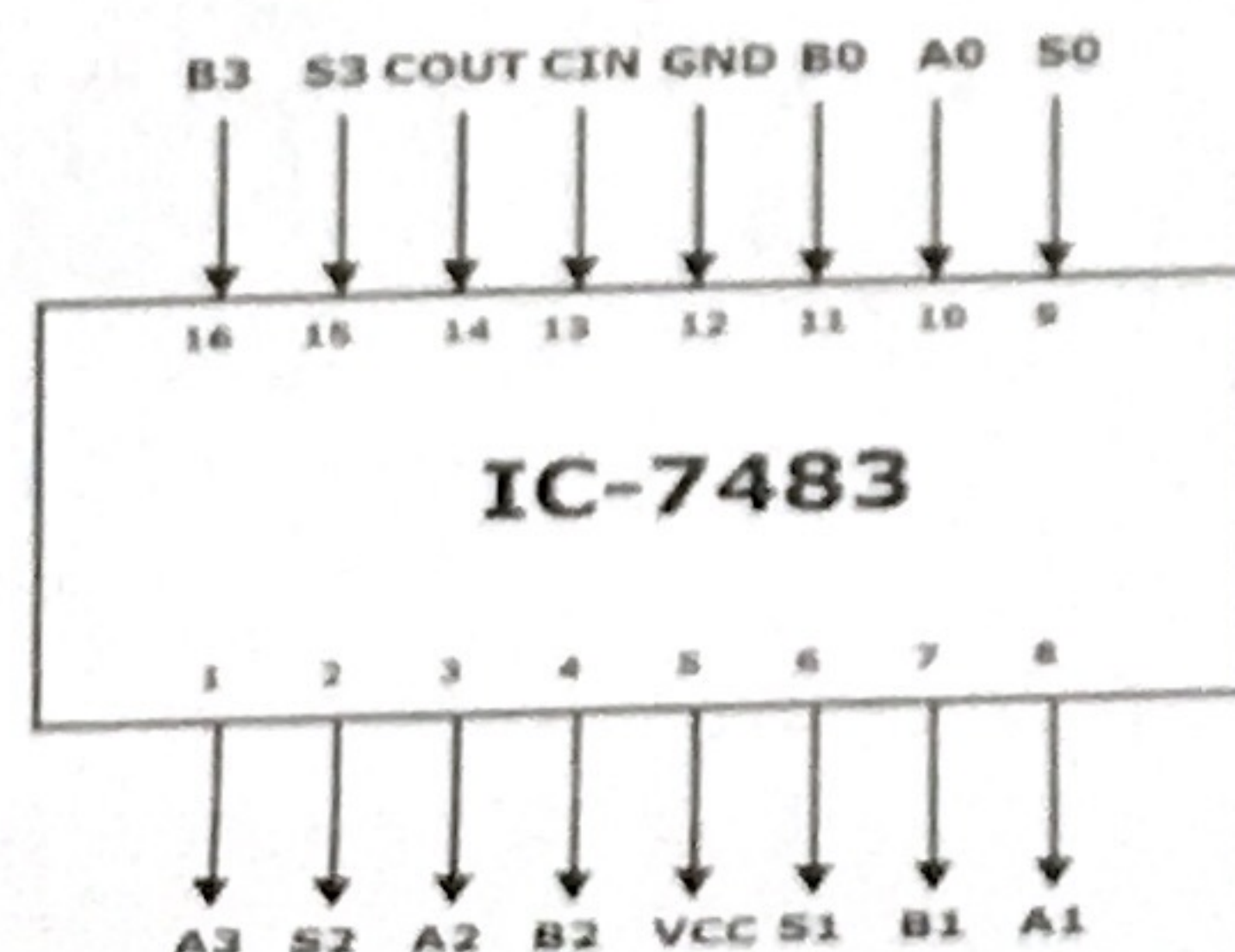
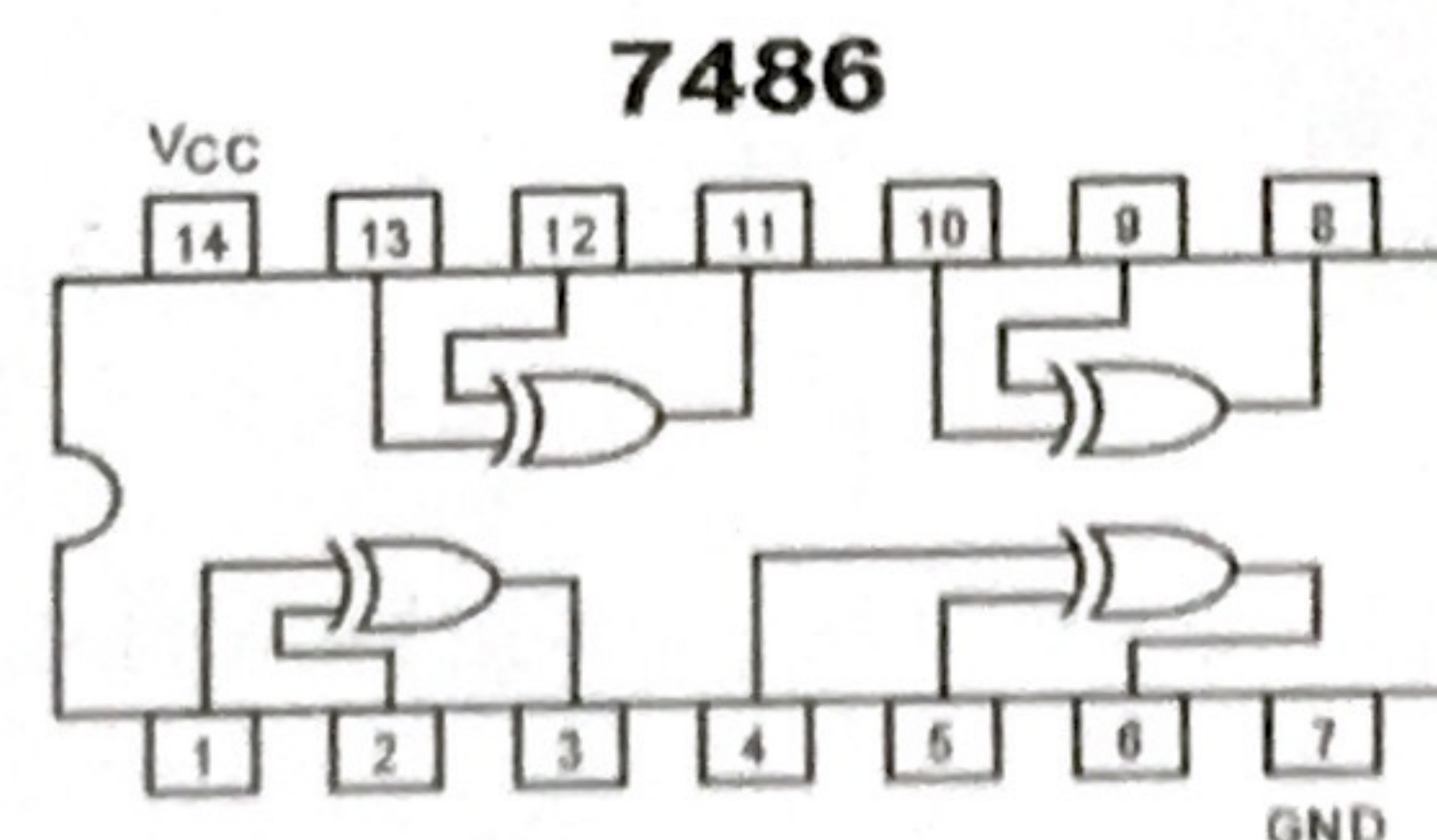
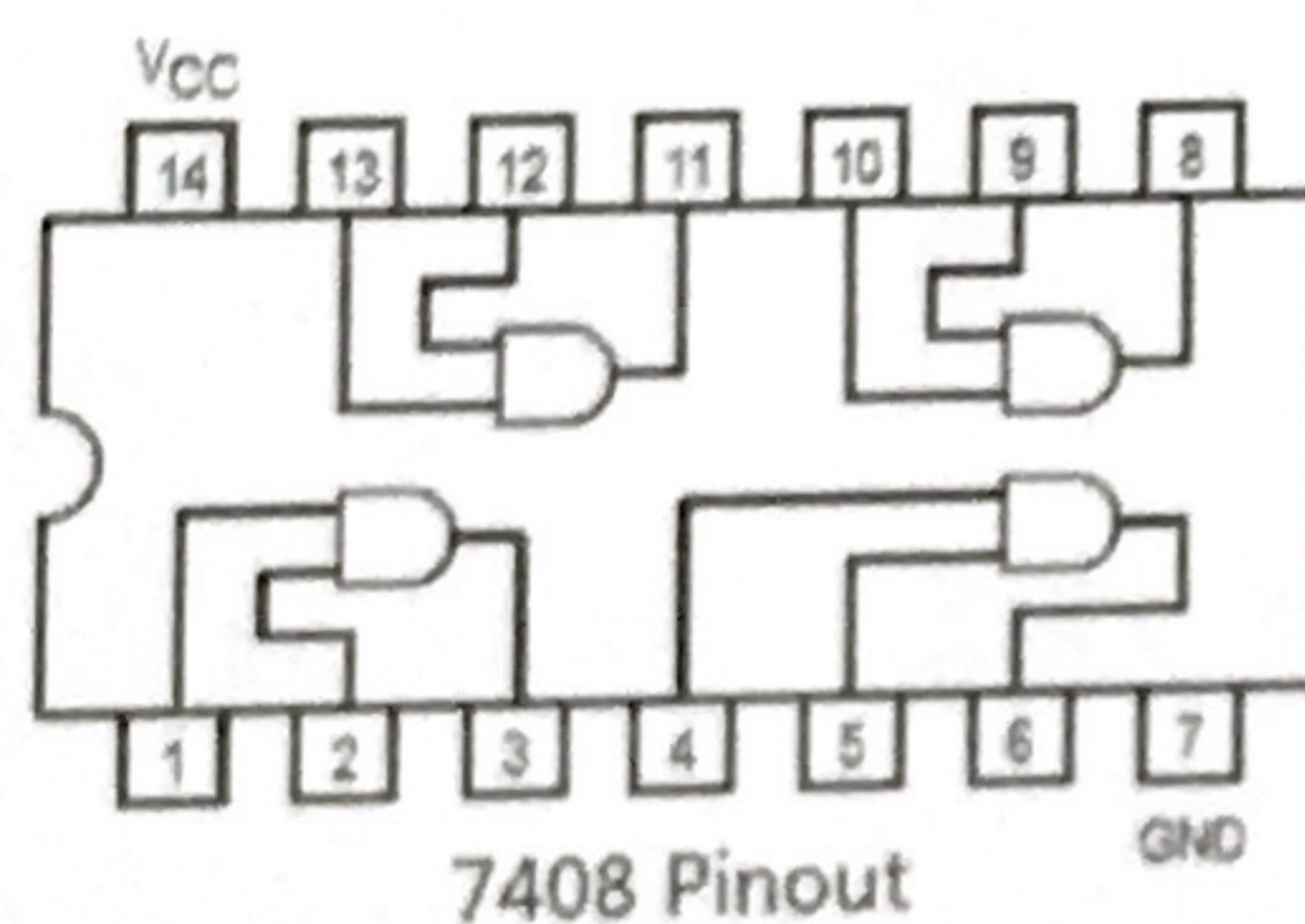
$$0 \leq P, Q \leq 3$$

The output must be presented in a 7-segment display module. Include necessary equations, truth tables and illustrate the circuit diagram in the script.

You will be allowed to use the following set of ICs for implementation (both hardware and Proteus).

- IC-7408 (2 input AND)
- IC-7483 (4 Bit Binary Parallel Adder)
- IC-7486 (2 input XOR)
- IC-74151 (8x1 Mux)

The Pin diagram of the referred components are illustrated below for convenience.



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SET-B

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INSTRUCTIONS:

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Marks

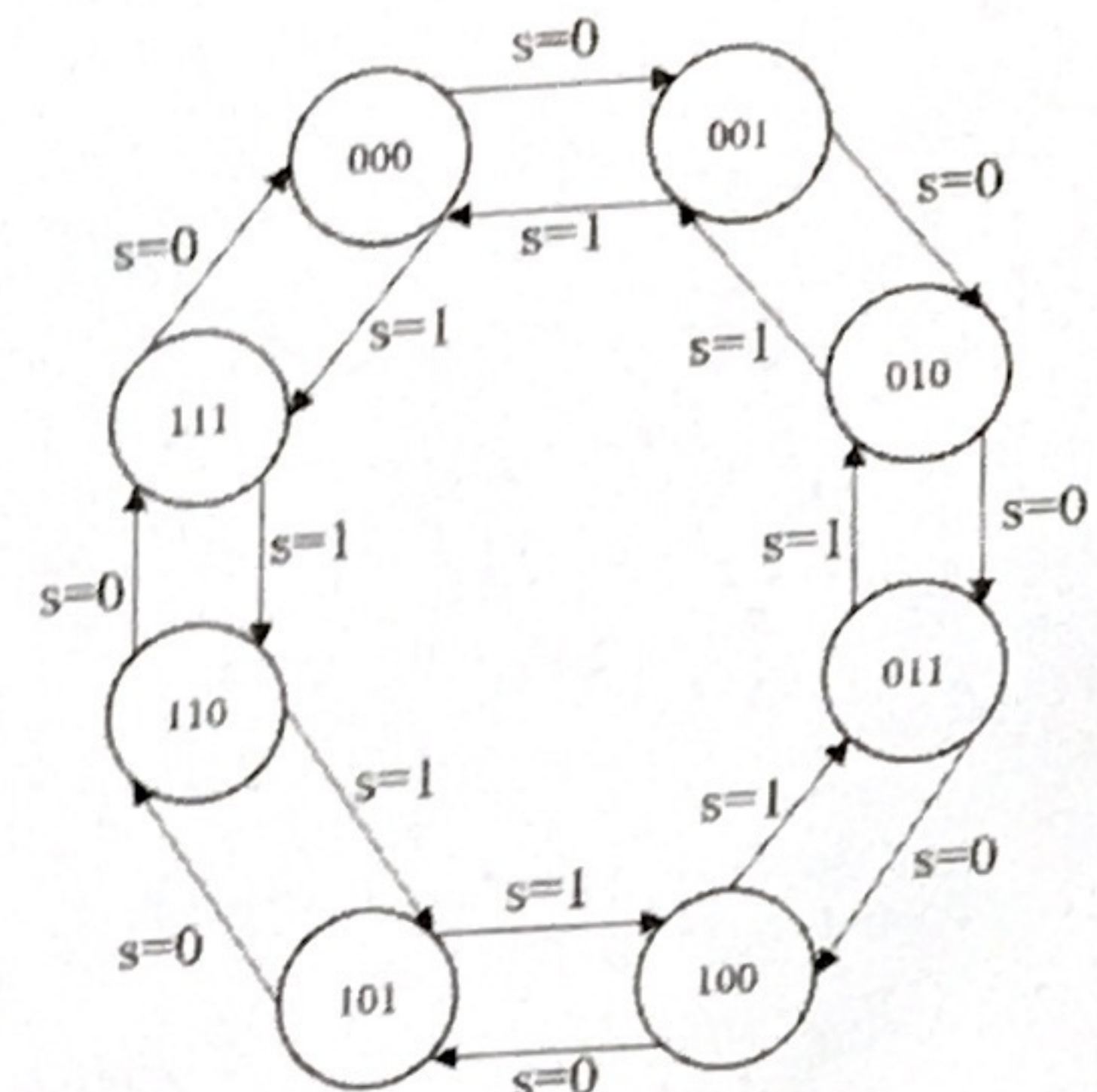
Question-1

20

Consider a 3-bit binary counter that works based on the following principle.

$$A(t+1) = \begin{cases} A(t) + 1, & s = 0 \\ A(t) - 1, & s = 1 \end{cases}$$

Here s works as a selection bit and the states of the counter have been represented by A . The state diagram of the counter is illustrated at right.



The states of the counter must be shown using a 7-segment display module. Include necessary equations, truth tables, excitation tables, K-maps and illustrate the circuit diagram in the script.

You will be allowed to use the following set of ICs for implementation (both hardware and Proteus).

- IC-7408 (2 input AND)
- IC-7432 (2 input OR)
- IC-7486 (2 input XOR)
- IC-74112 (J-K Flip Flop)
[For Proteus implementation it is allowed to use IC-74111 for a J-K Flip Flop]

The Pin diagram of the referred components are illustrated below for convenience.

